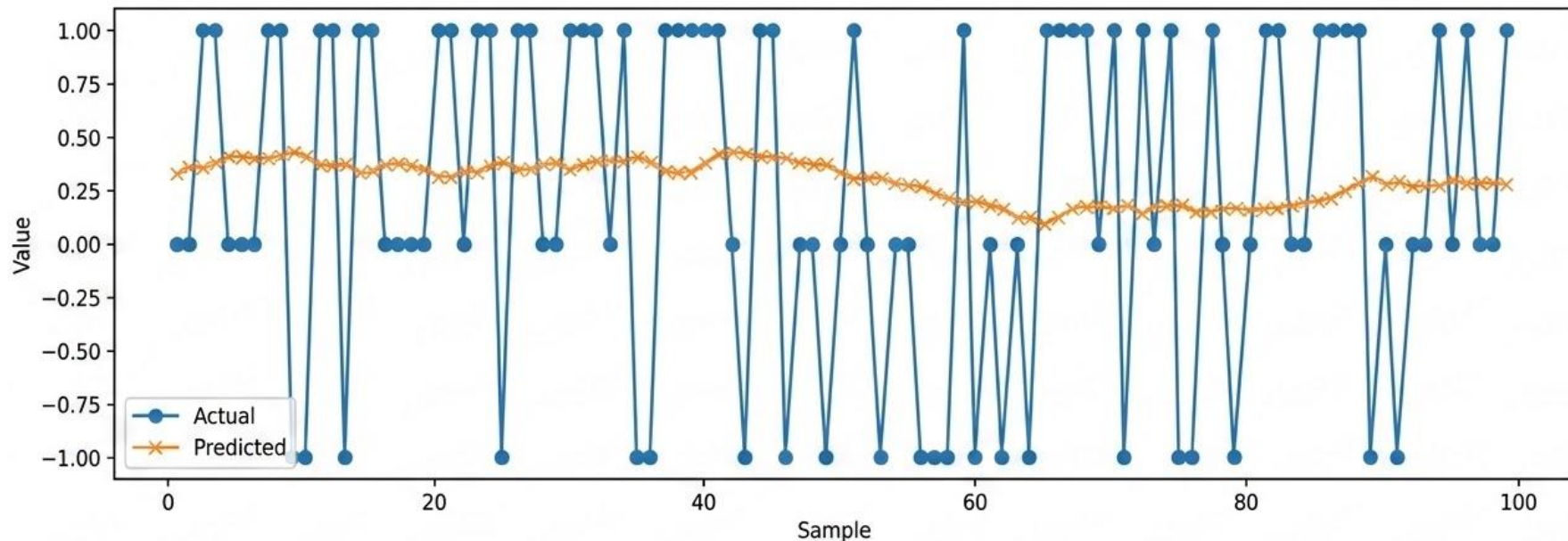


Tesla Stock

Neural Network Predictions vs Actual



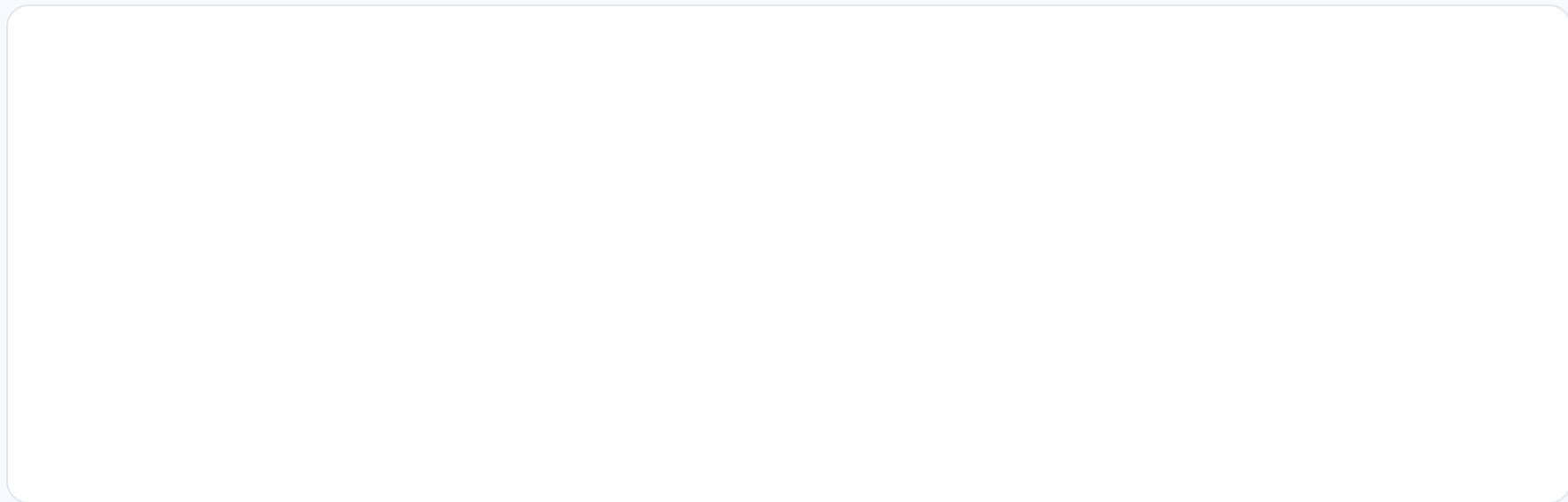
Validation score:
0.903584

Train R^2 Score:
0.9064

Test R^2 Score:
0.9023

SVM Performance Analysis

Loss vs Accuracy curve



Accuracy

0.6972

F1

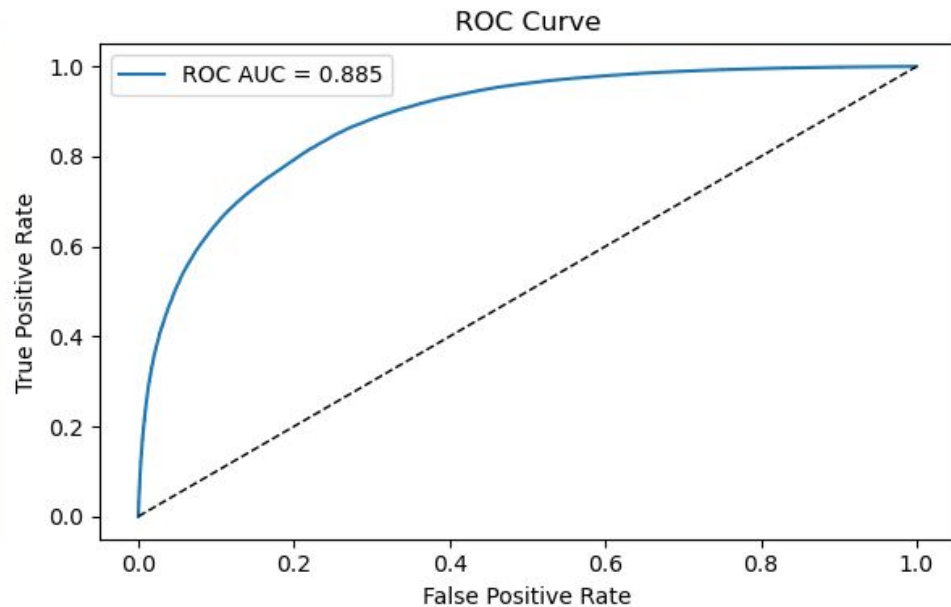
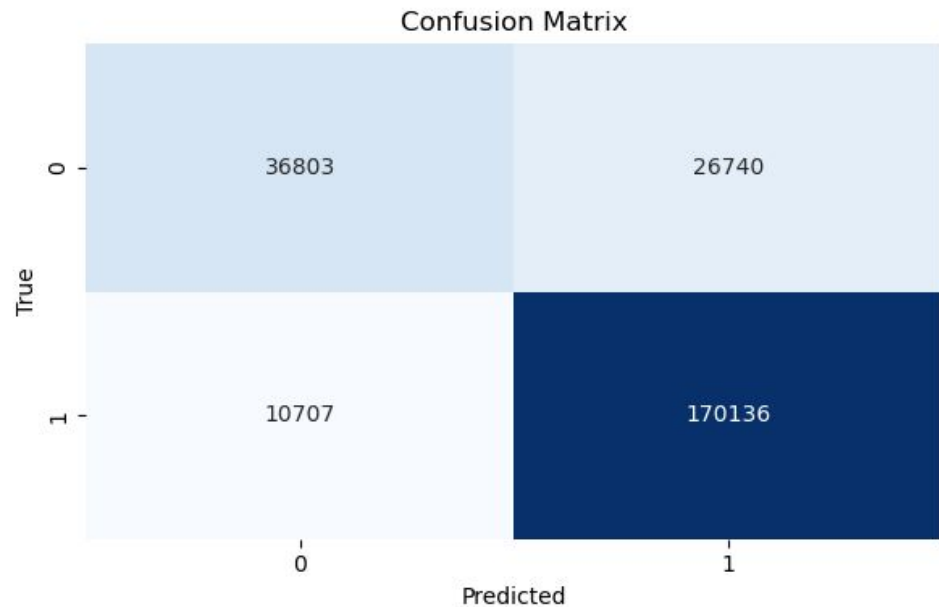
0.6969

MCC

0.3951

Linear Regression Performance Analysis

Loss vs Accuracy curve



Accuracy Score

0.8468

Class 0 F1-score

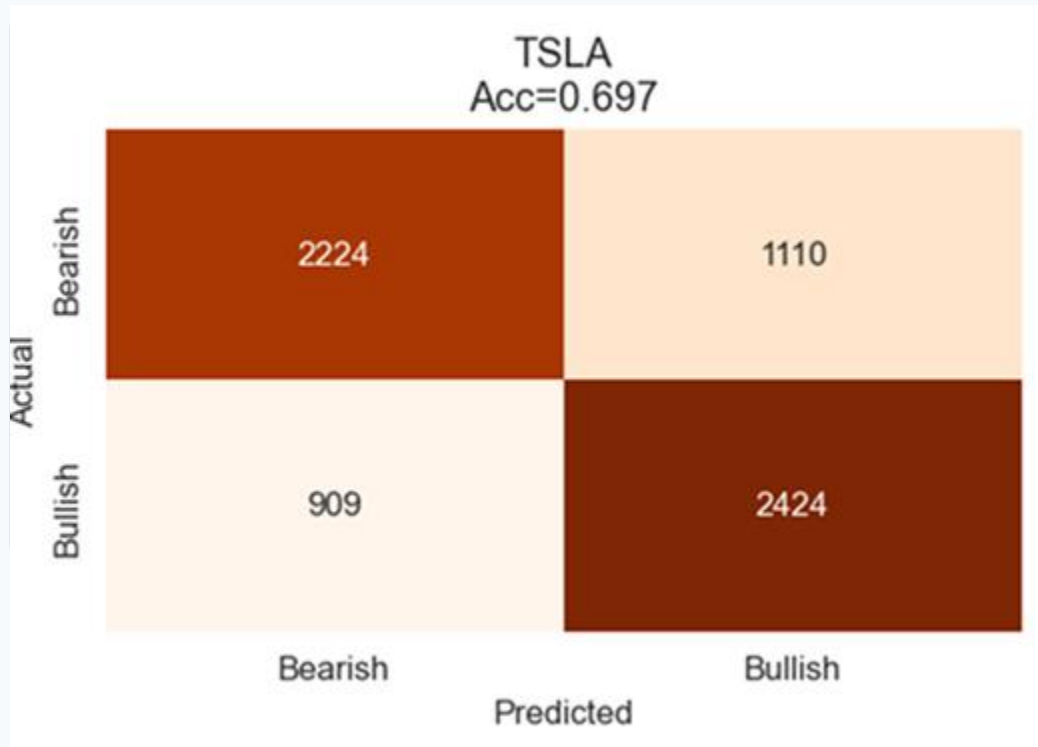
0.6889

Class 1 F1-score

0.8888

Binary Confusion Matrix

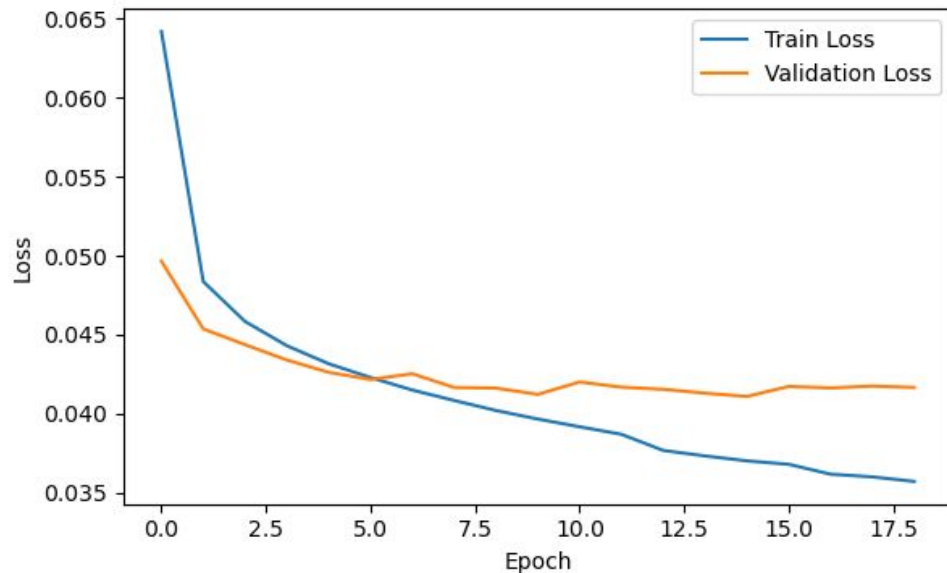
Bullish vs Bearish



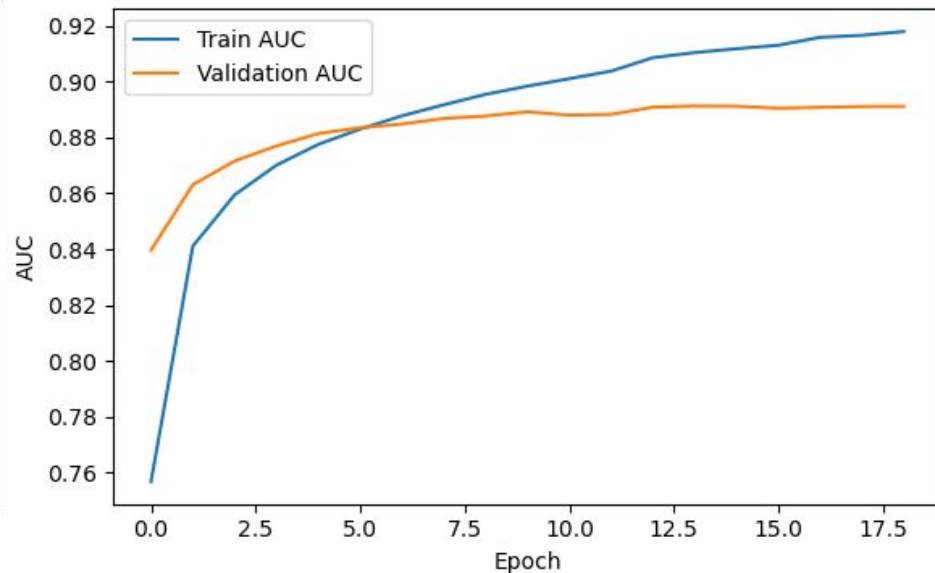
LSTM Performance Analysis

Loss vs AUC

Loss Curve



AUC Curve



Accuracy Score

0.8412

Class 0 F1-score

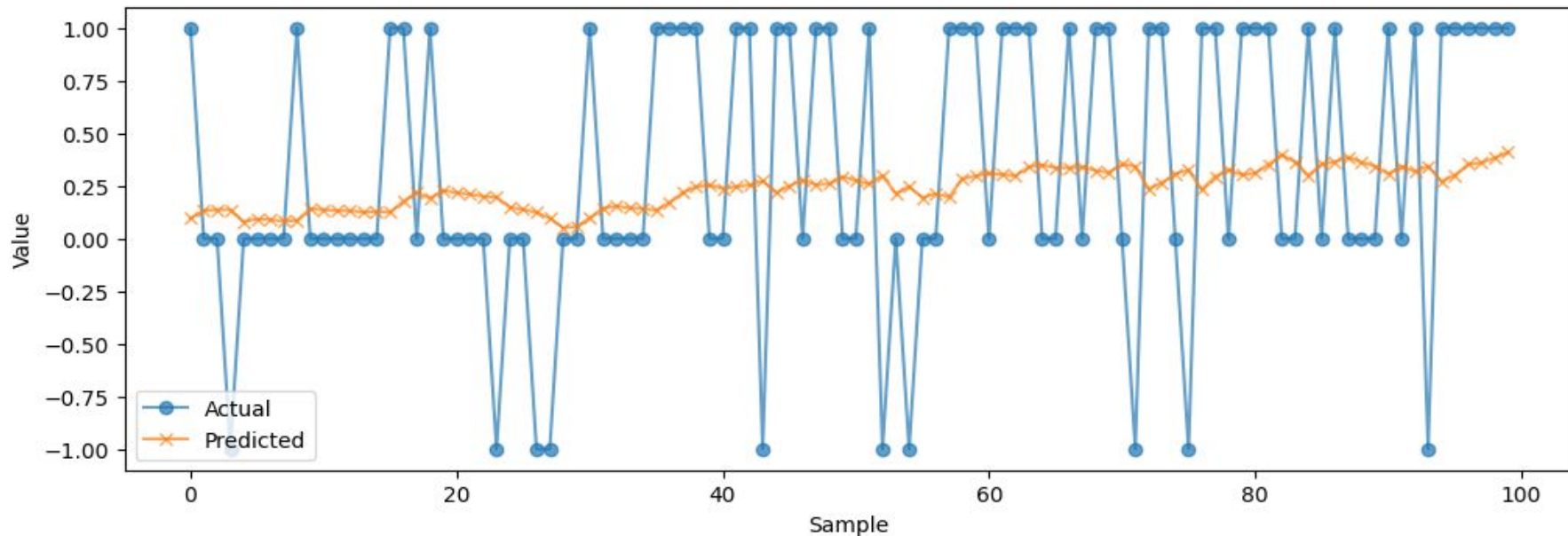
0.6953

Class 1 F1-score

0.8926

Apple Stock

Neural Network Predictions vs Actual



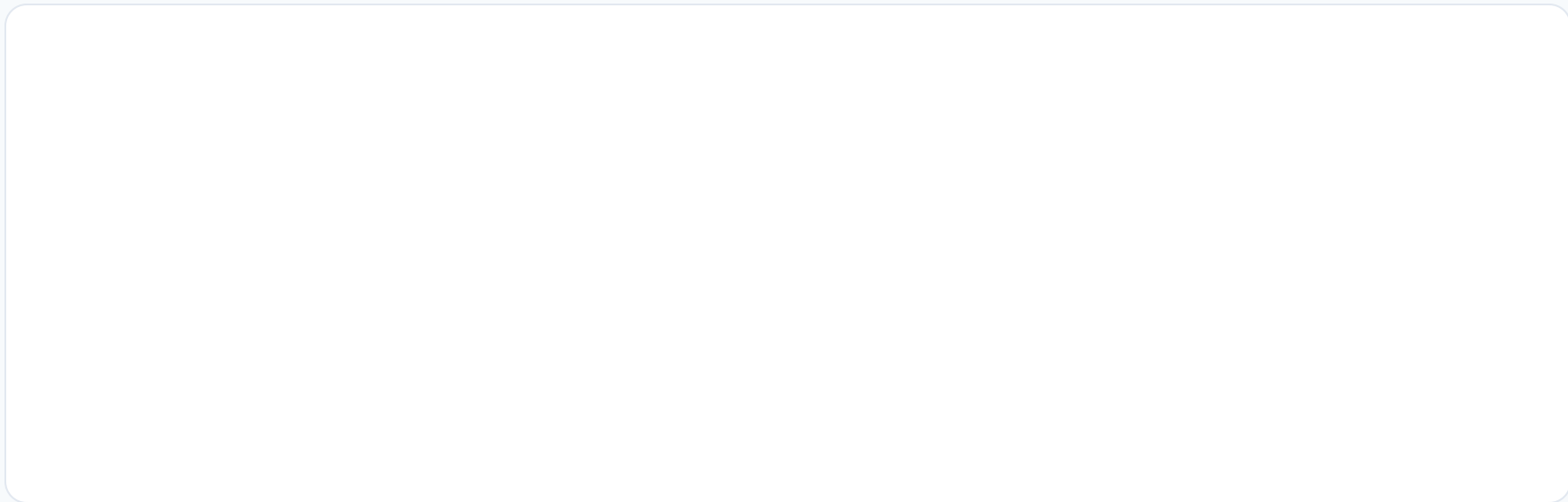
Validation score:
0.899173

Train R^2 Score:
0.9001

Test R^2 Score:
0.8910

SVM Performance Analysis

Loss vs Accuracy curve



Accuracy Score

0.7266

F1 Score

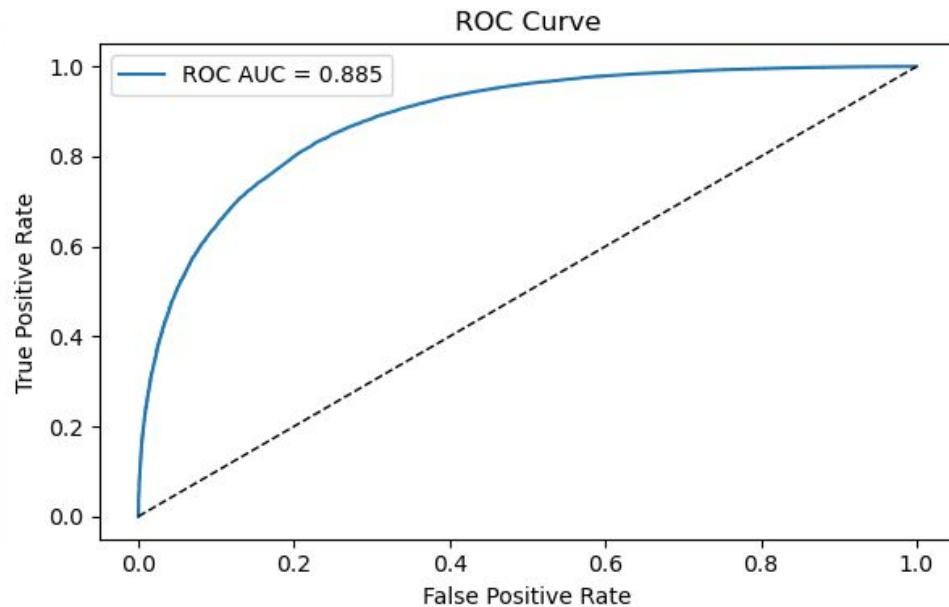
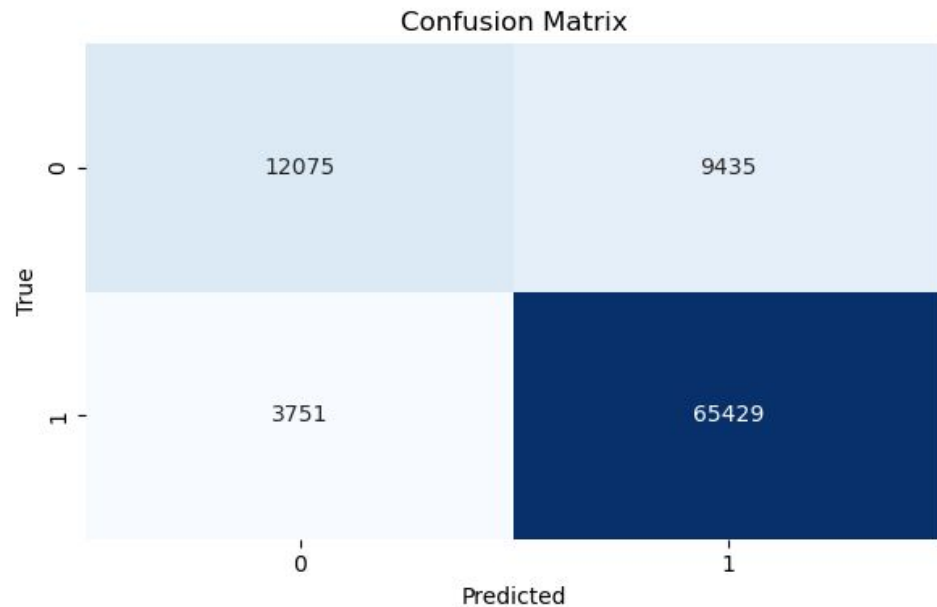
0.7263

MCC

0.4541

Linear Regression Performance Analysis

Loss vs Accuracy curve



Accuracy Score

0.8546

Class 0 F1-score

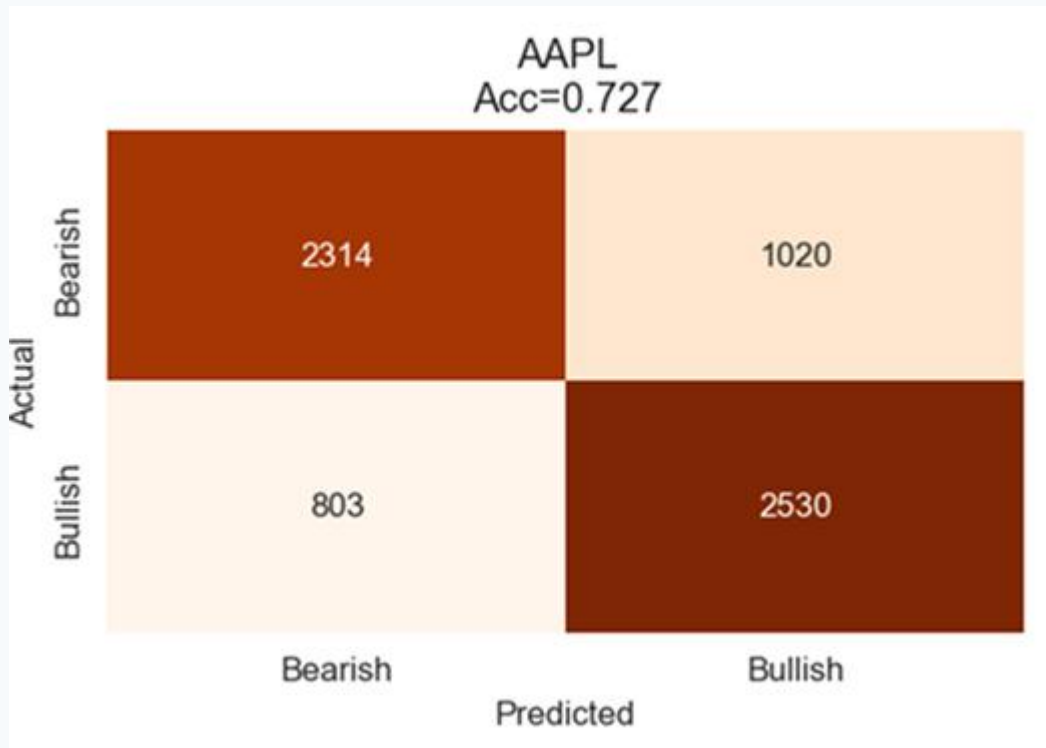
0.6765

Class 1 F1-score

0.8959

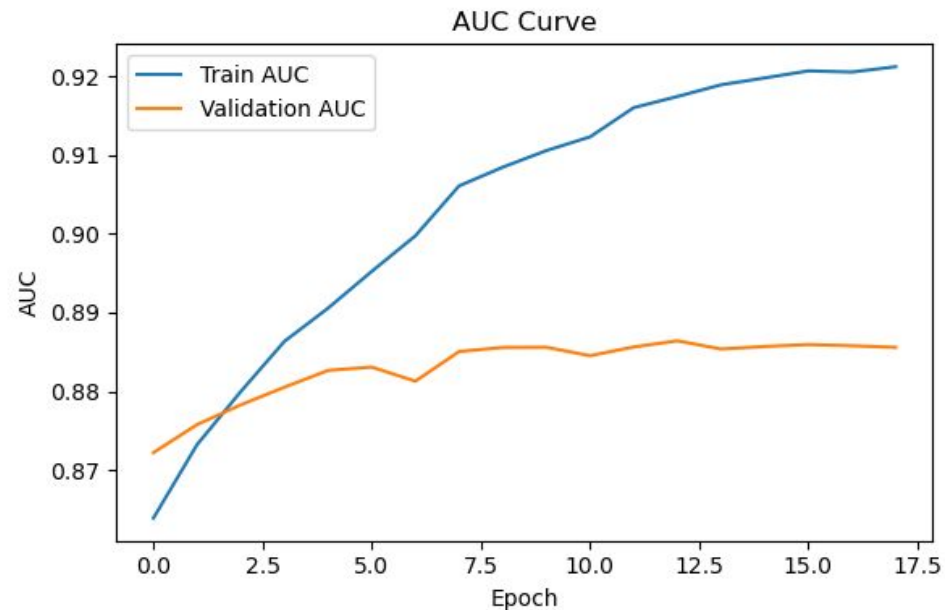
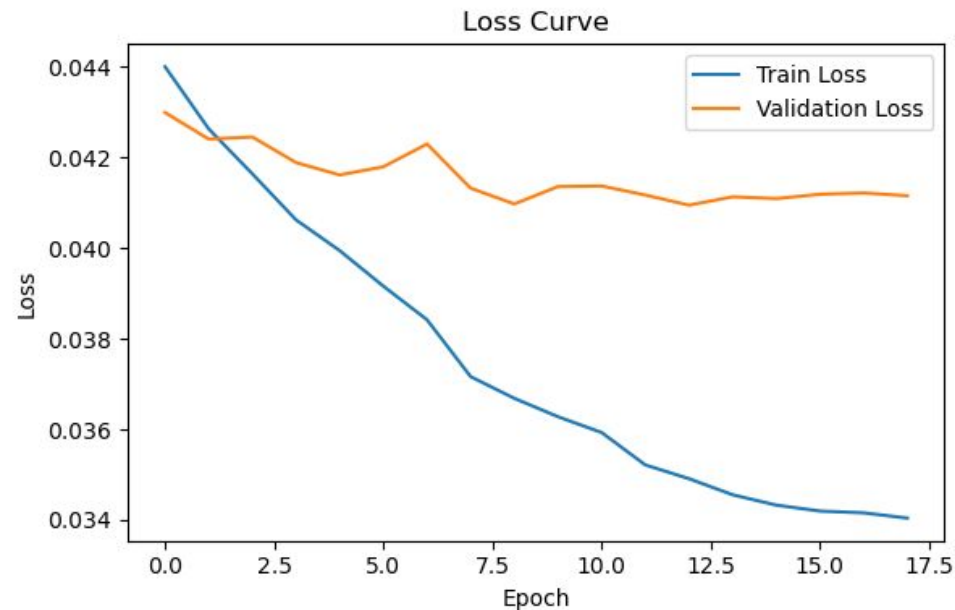
Binary Confusion Matrix

Bullish vs Bearish



LSTM Performance Analysis

Loss vs AUC



Accuracy Score

0.8452

Class 0 F1-score

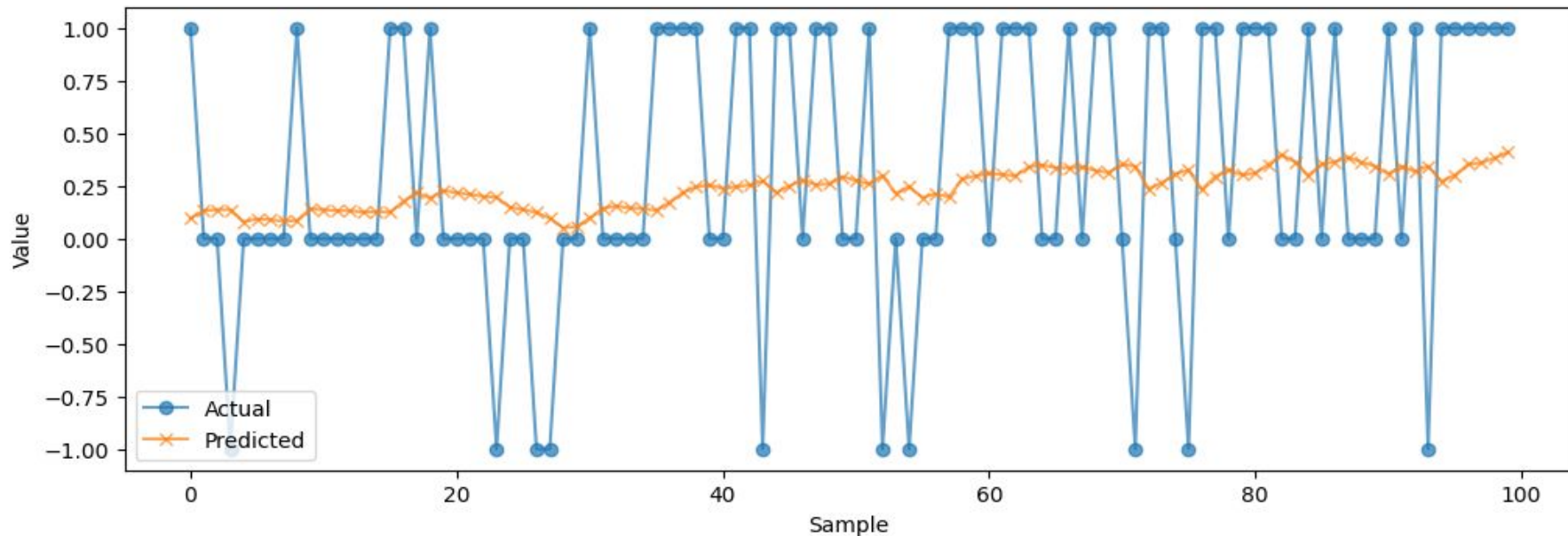
0.6798

Class 1 F1-score

0.8980

Amazon Stock

Neural Network Predictions vs Actual



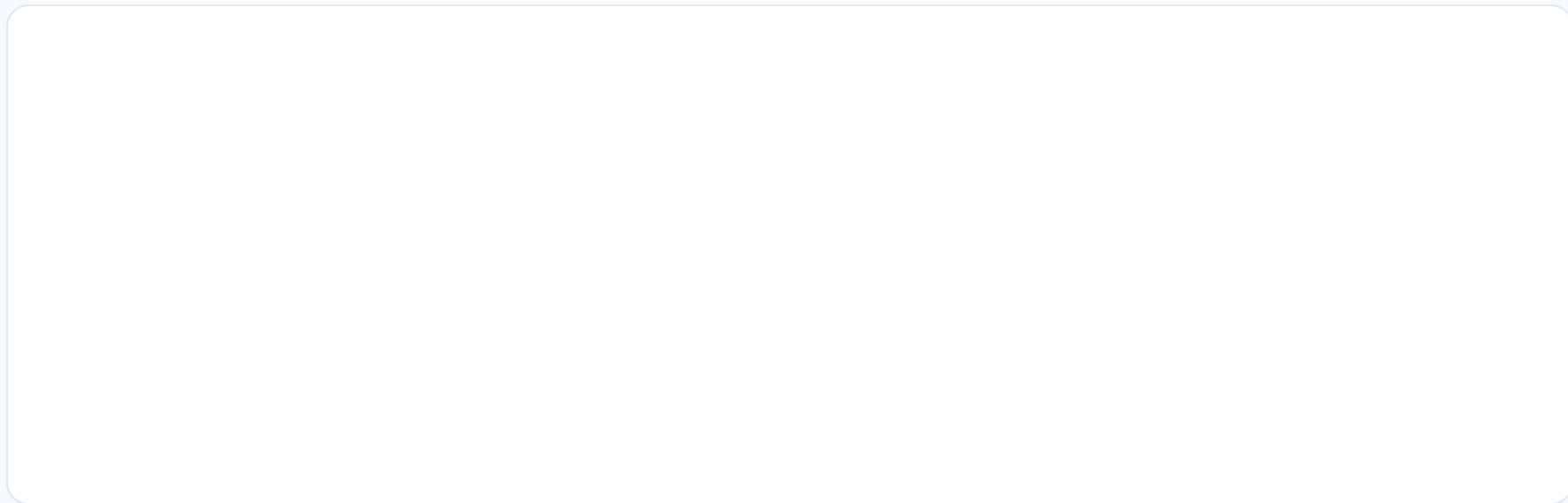
Validation score:
0.880993

Train R^2 Score:
0.8860

Test R^2 Score:
0.8817

SVM Performance Analysis

Loss vs Accuracy curve



Accuracy Score

0.7231

F1 Score

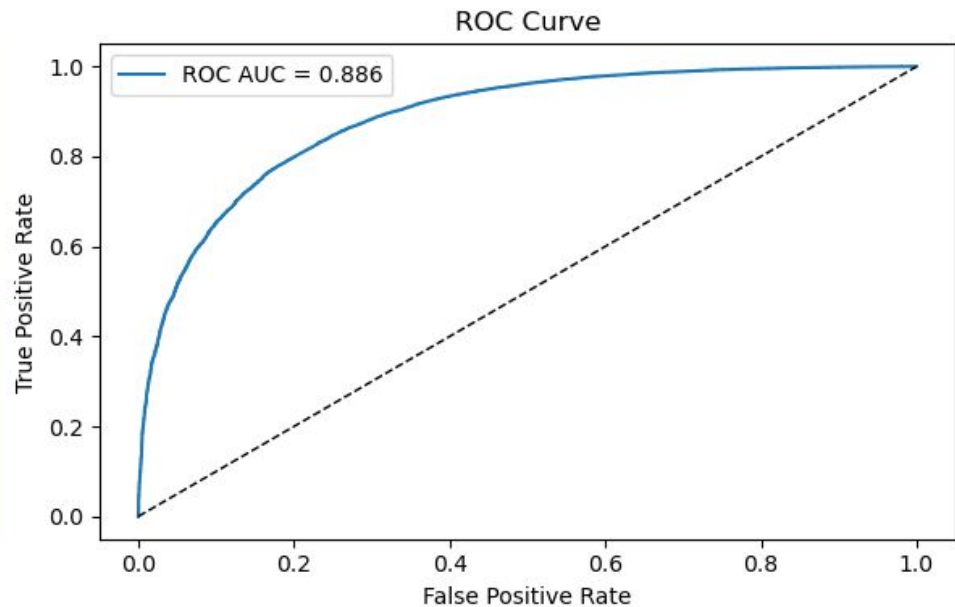
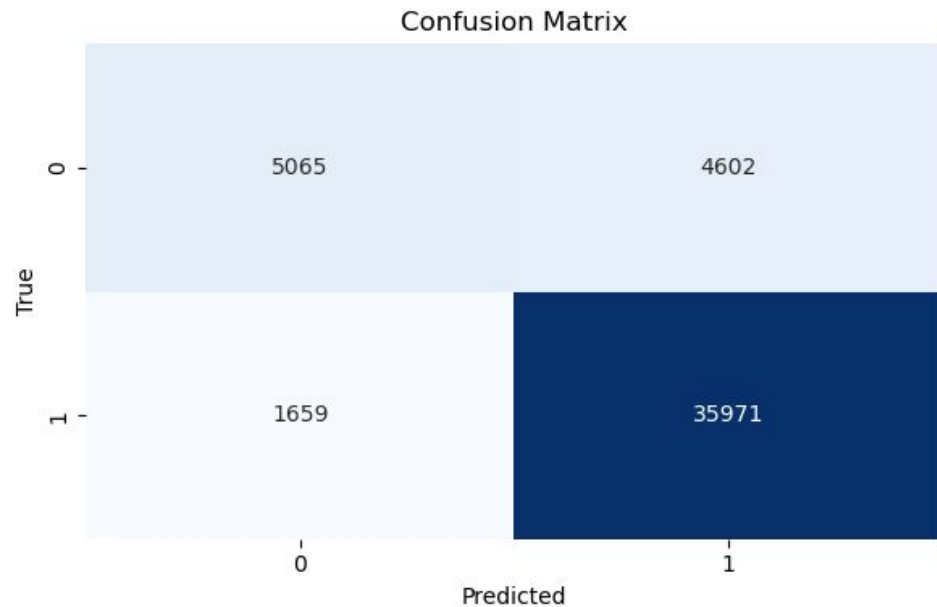
0.7231

MCC

0.4464

Linear Regression Performance Analysis

Loss vs Accuracy curve



Accuracy Score

0.8676

Class 0 F1-score

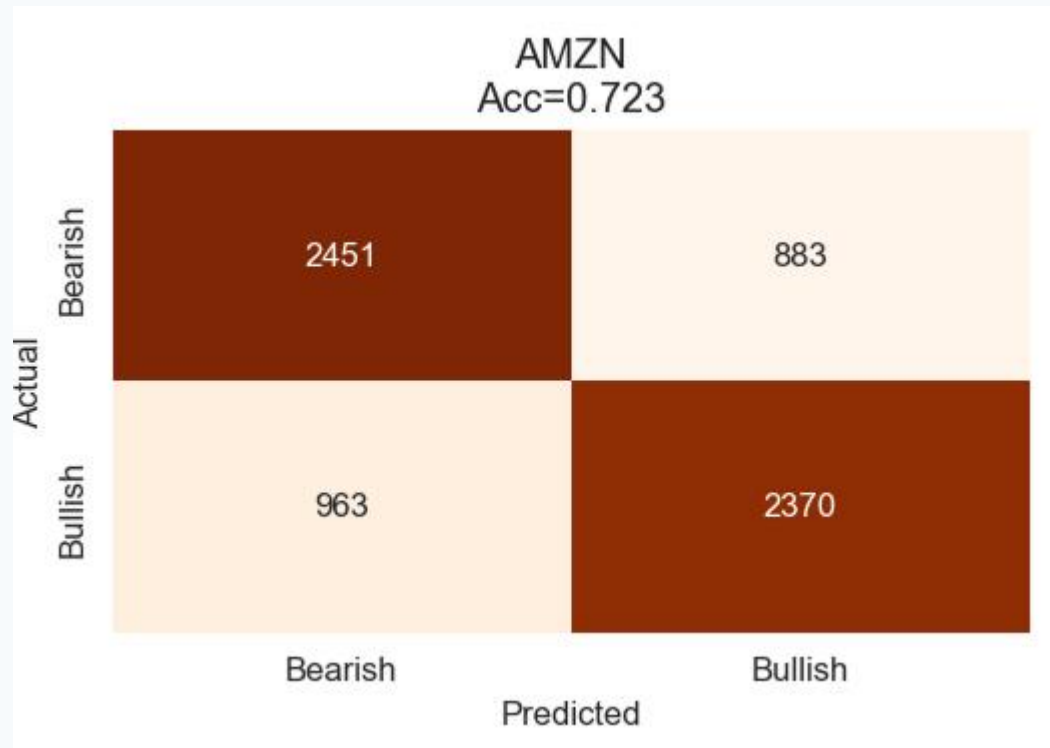
0.6522

Class 1 F1-score

0.9125

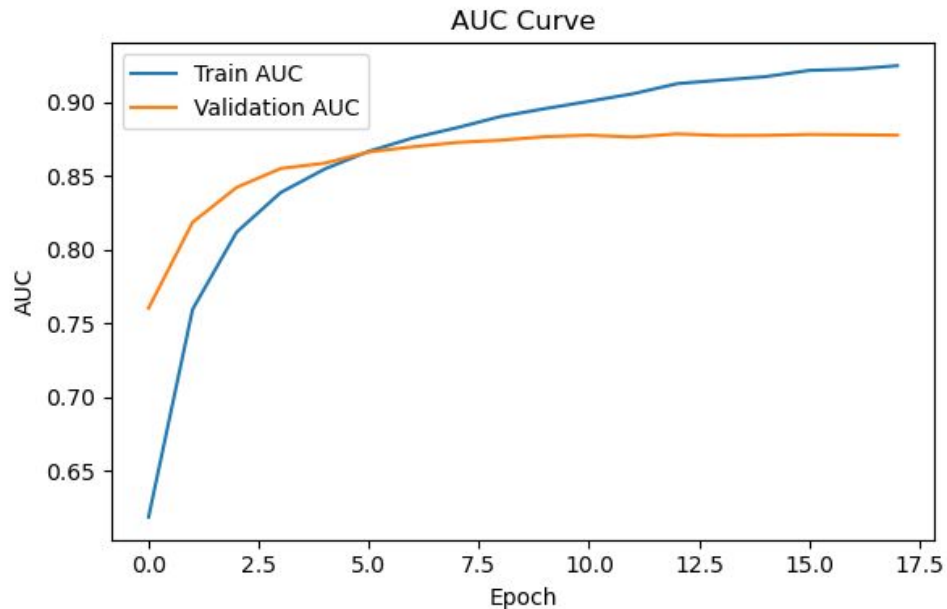
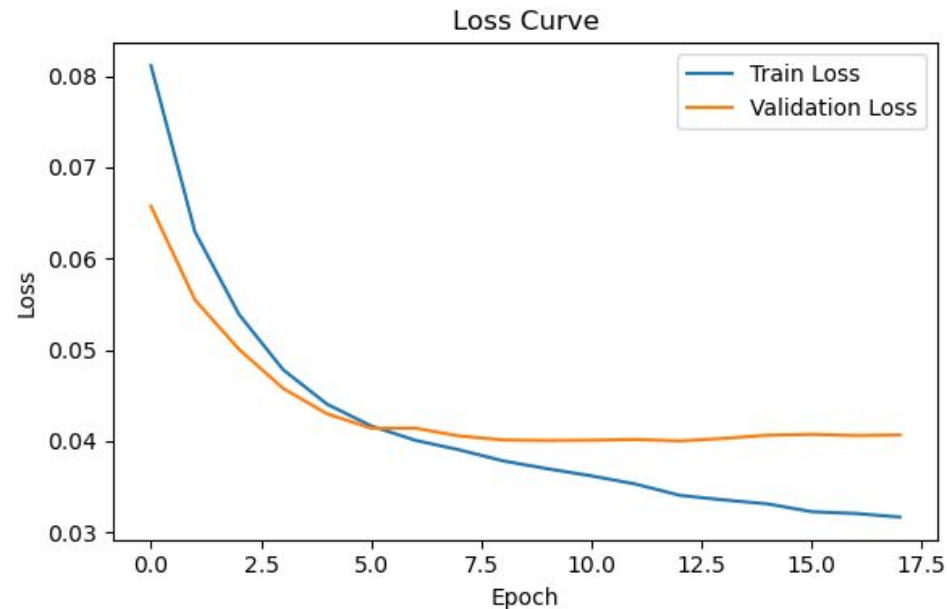
Binary Confusion Matrix

Bullish vs Bearish



LSTM Performance Analysis

Loss vs AUC



Accuracy Score

0.8522

Class 0 F1-score

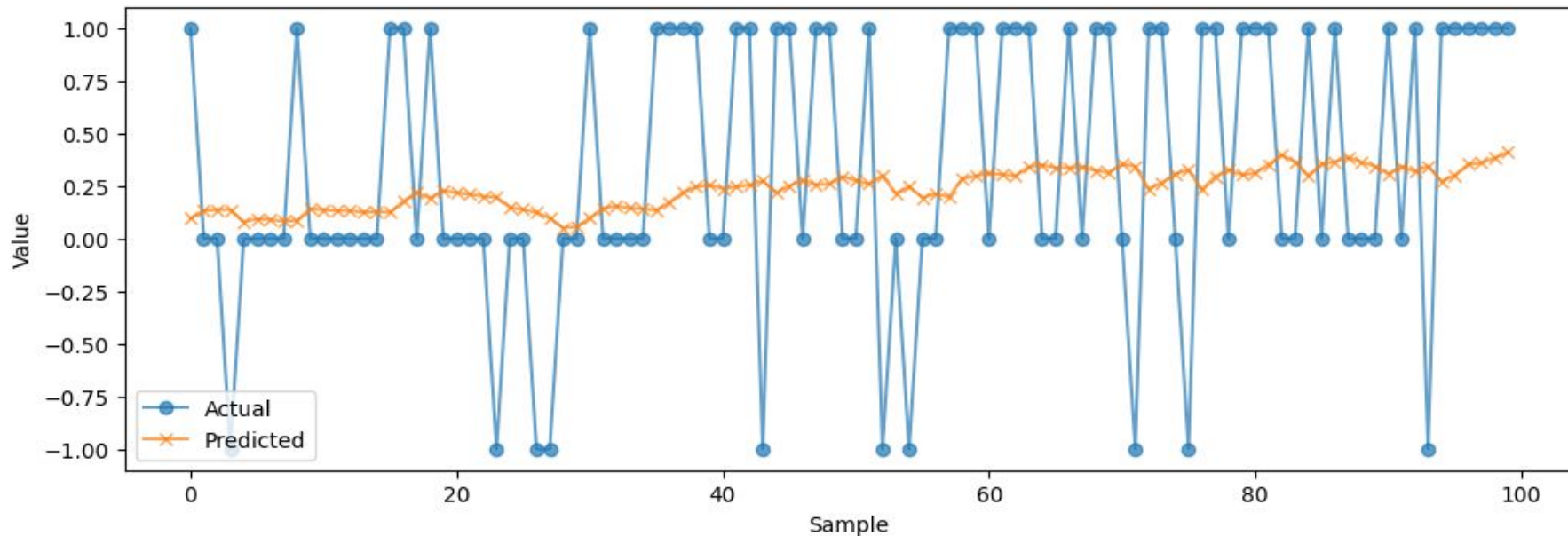
0.6382

Class 1 F1-score

0.9071

Nvidia Stock

Neural Network Predictions vs Actual



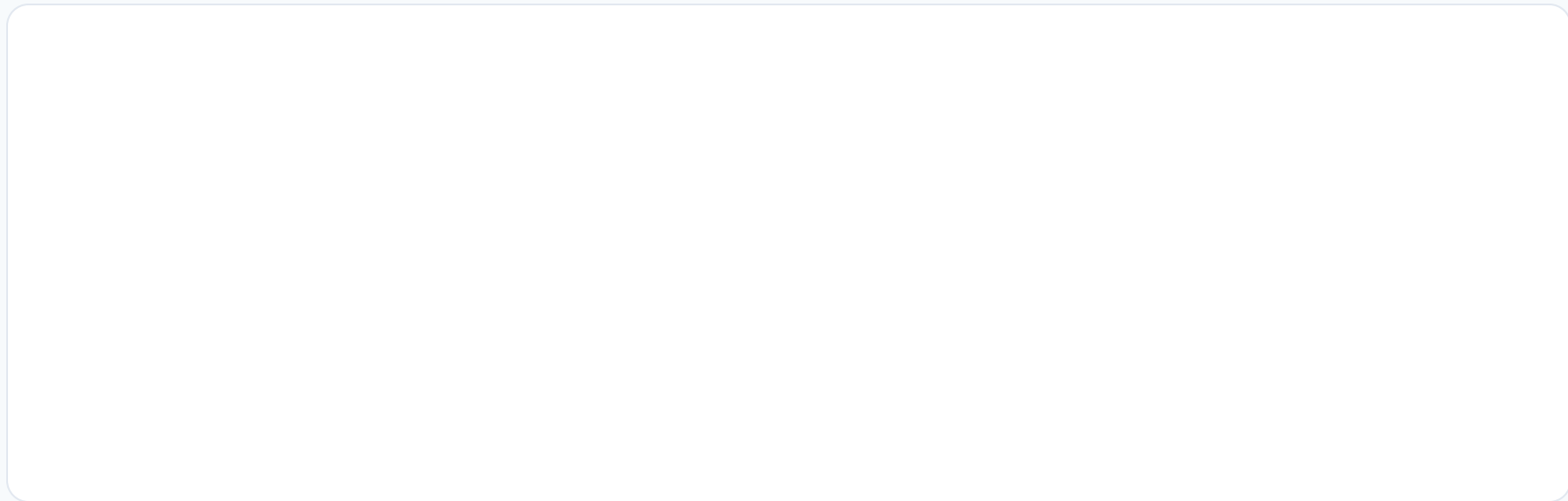
Validation score:
0.869210

Train R^2 Score:
0.8715

Test R^2 Score:
0.8809

SVM Performance Analysis

Loss vs Accuracy curve



Accuracy Score

0.7315

F1 Score

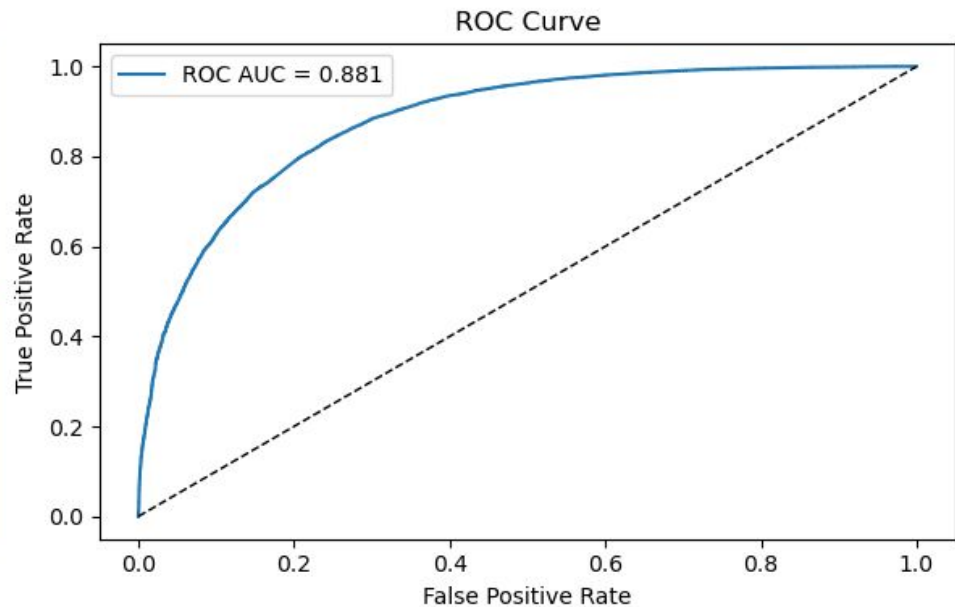
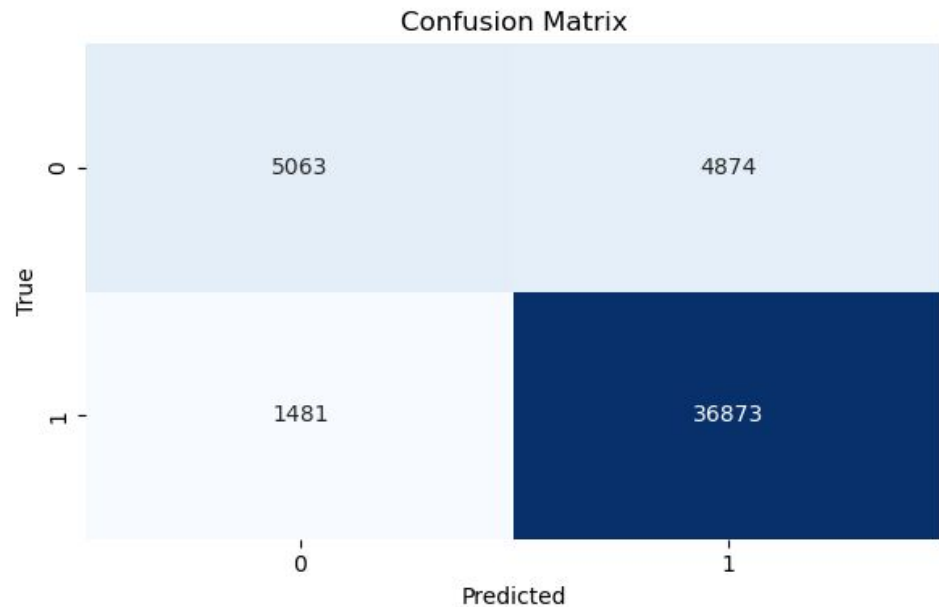
0.7313

MCC

0.4639

Linear Regression Performance Analysis

Loss vs Accuracy curve



Accuracy Score

0.8684

Class 0 F1-score

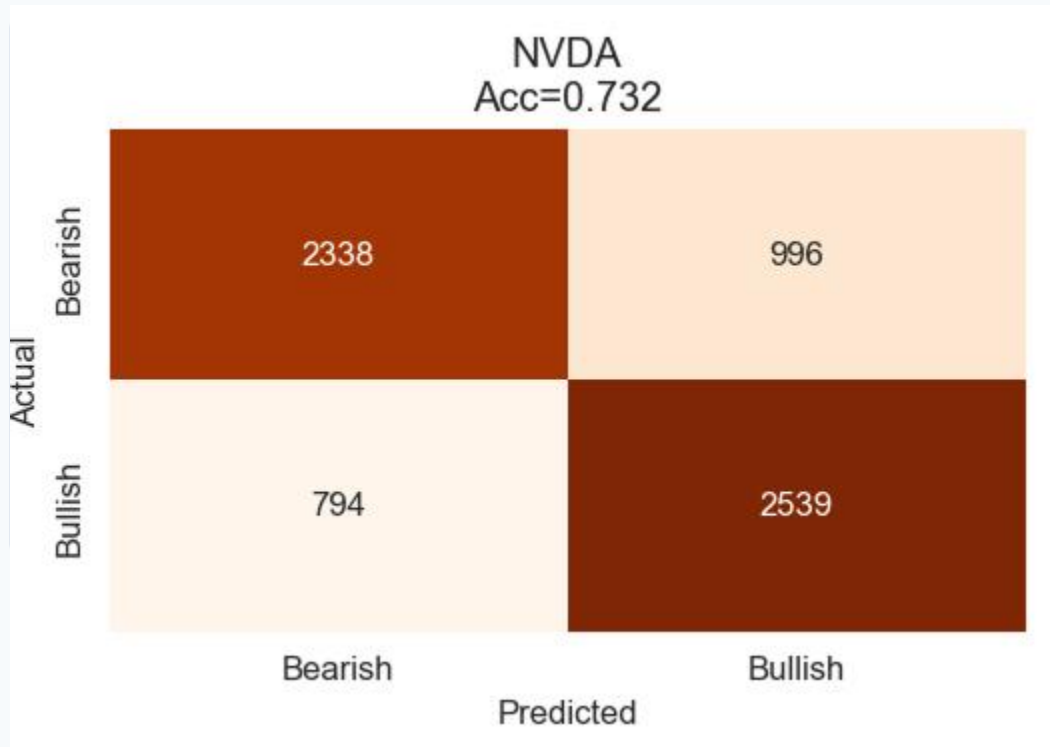
0.6535

Class 1 F1-score

0.9128

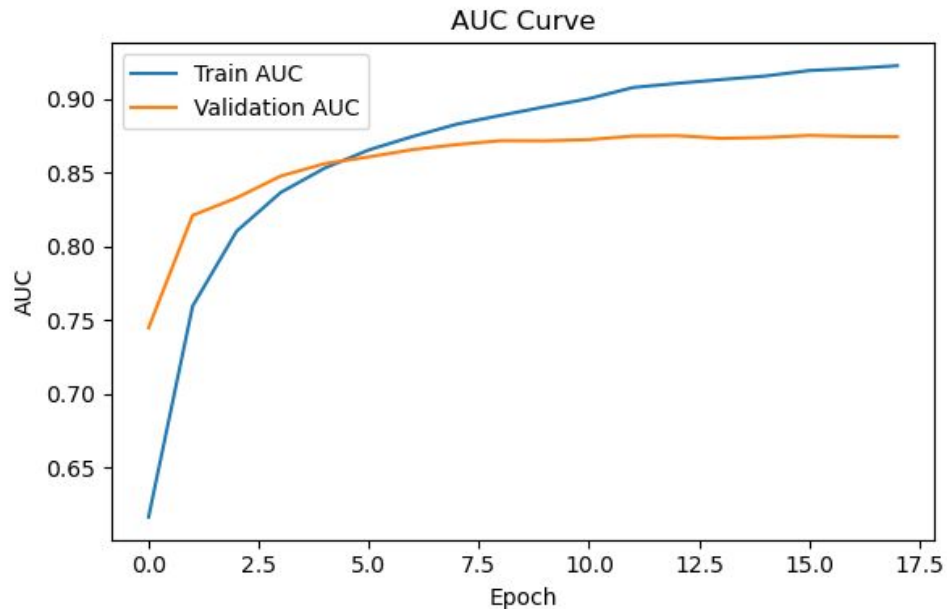
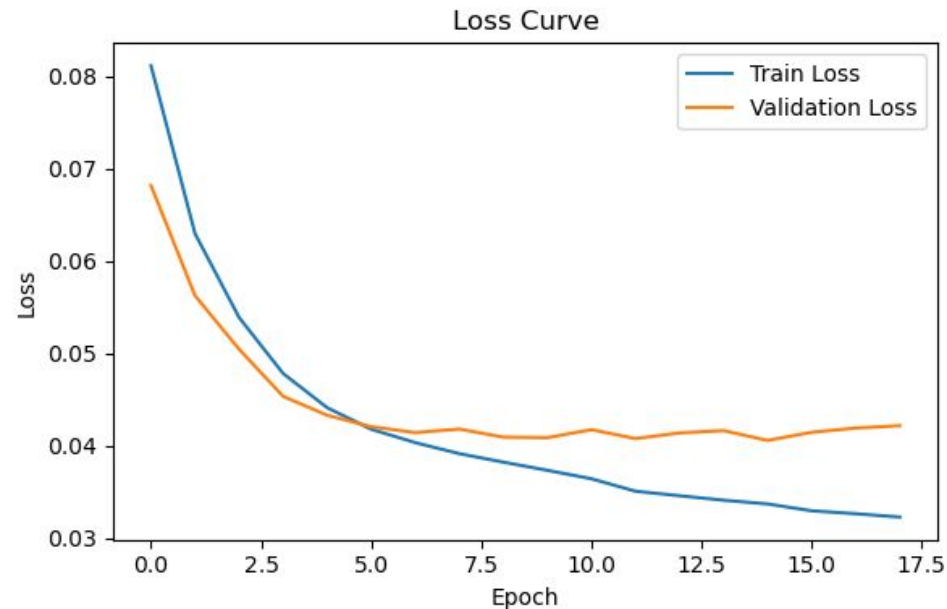
Binary Confusion Matrix

Bullish vs Bearish



LSTM Performance Analysis

Loss vs AUC



Accuracy Score

0.8452

Class 0 F1-score

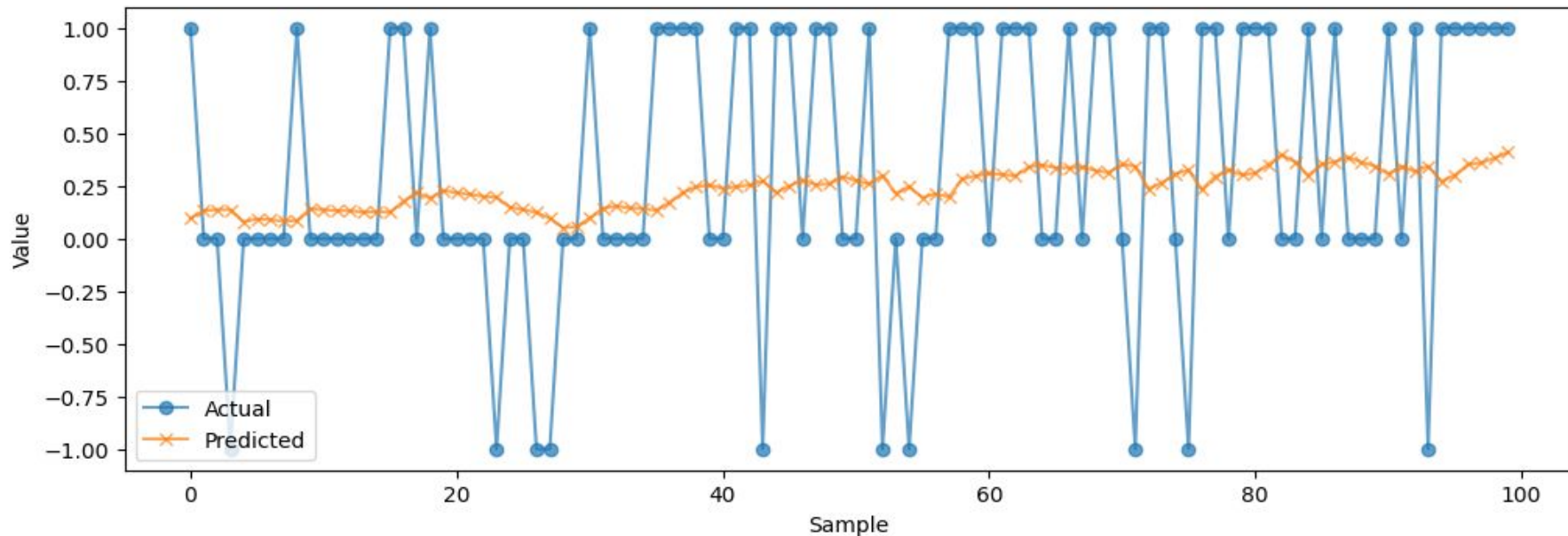
0.6798

Class 1 F1-score

0.8980

Meta Stock

Neural Network Predictions vs Actual



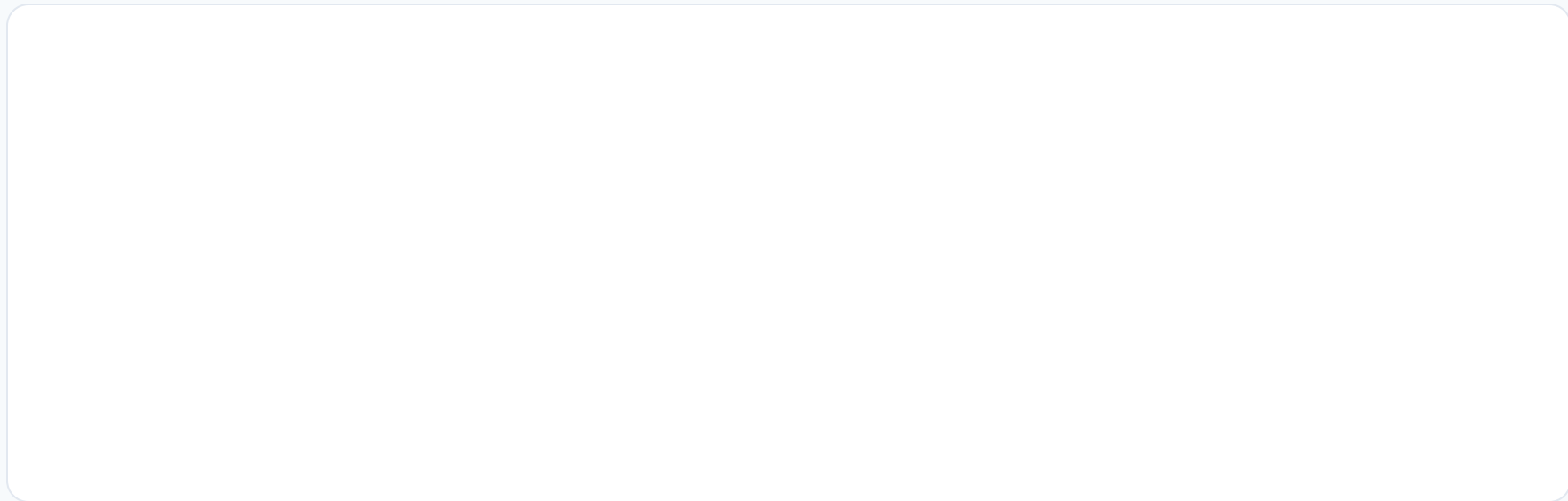
Validation score:
0.872681

Train R^2 Score:
0.8753

Test R^2 Score:
-1.8695

SVM Performance Analysis

Loss vs Accuracy curve



Accuracy Score

0.7497

F1 Score

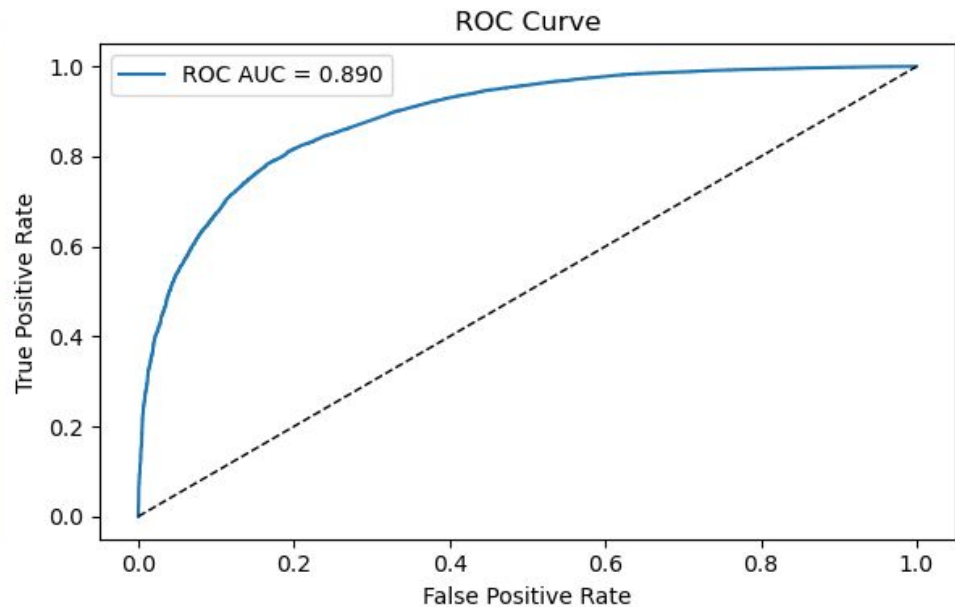
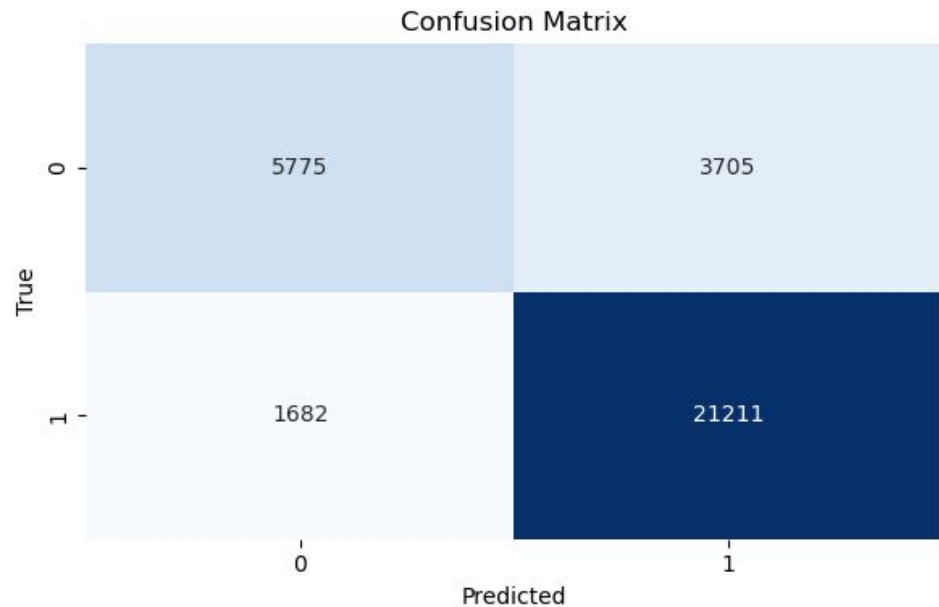
0.7496

MCC

0.4996

Linear Regression Performance Analysis

Loss vs Accuracy curve



Accuracy Score

0.8336

Class 0 F1-score

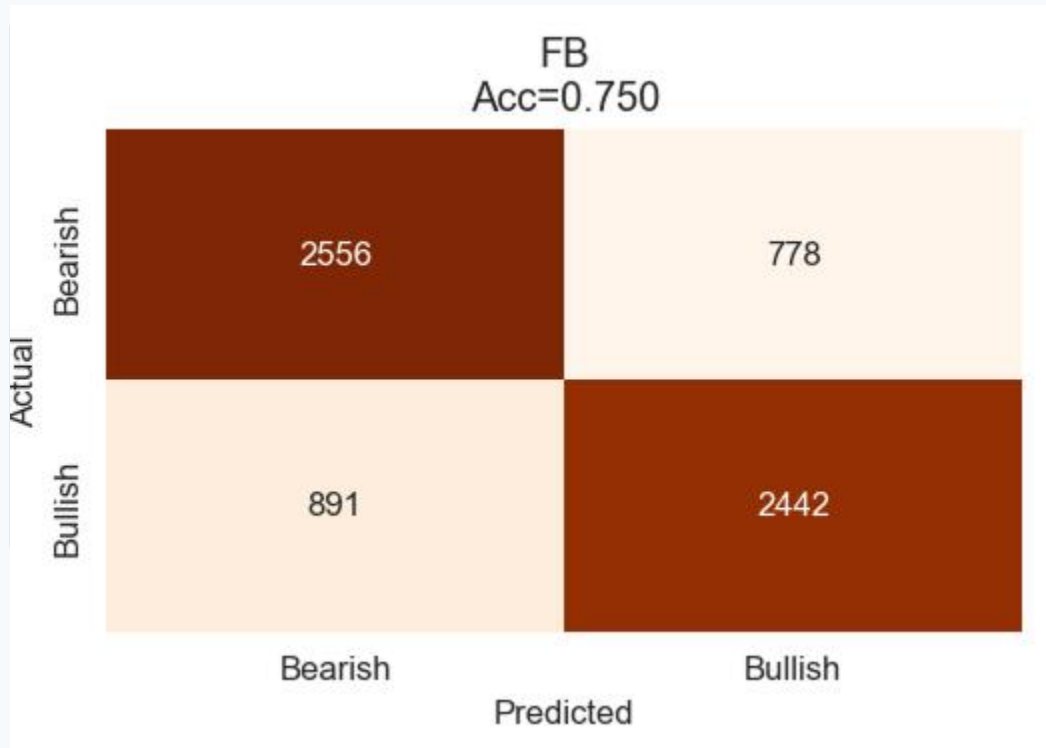
0.7141

Class 1 F1-score

0.8628

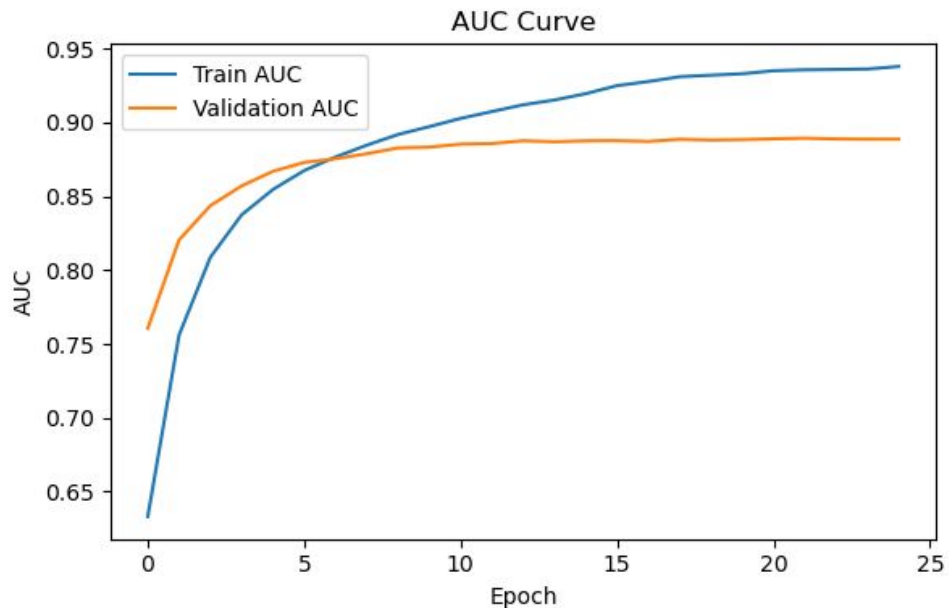
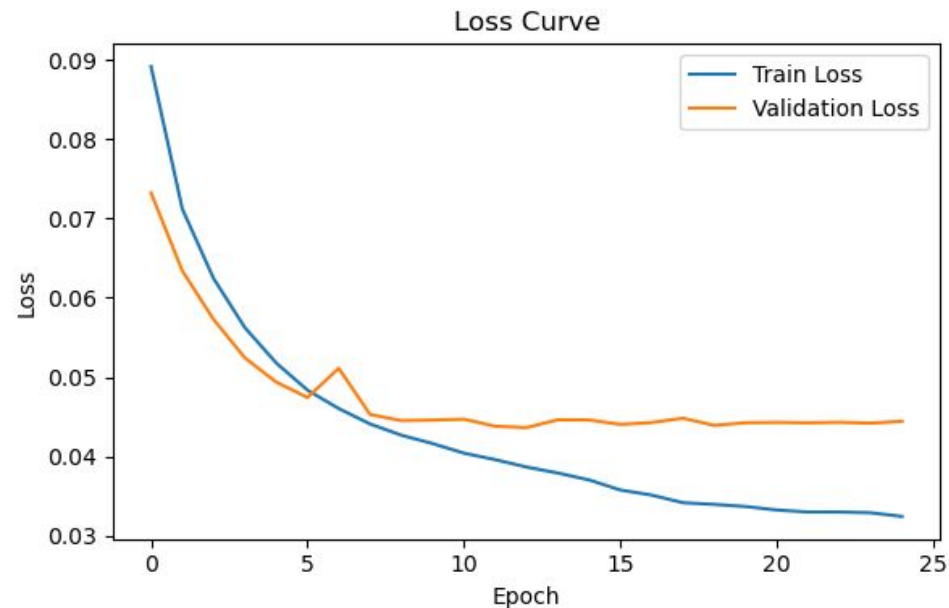
Binary Confusion Matrix

Bullish vs Bearish



LSTM Performance Analysis

Loss vs AUC



Accuracy Score

0.8155

Class 0 F1-score

0.7110

Class 1 F1-score

0.8645

Preprocessing

Data Acquisition & Initial Scope

Objective: Consolidate raw data for sentiment analysis

Data Source

Loaded cleaned ticker datasets from CSV files.

Initial Action

Filtered datasets to keep only relevant fields by dropping unneeded metadata.

Stocks Included

AAPL

Apple

AMZN

Amazon

META

(FB)

NVDA

NVIDIA

TSLA

Tesla

Sentiment Label Objective

Transform categorical sentiment into a quantitative format for machine learning

Target Feature

A standardized numeric label for sentiment classification.

Mapping Logic

1

Bullish



0

Neutral / Other



-1

Bearish



Text Cleaning Pipeline

Objective: Normalize post text (body) to reduce noise and improve model consistency.

Transformation

body → body_clean

Refinement Steps



Case Normalization

Converted all text to lowercase for uniformity.



Noise Removal

Stripped URLs, HTML tags, emojis, @handles, and #hashtags.



Standardization

Normalized whitespace and expanded contractions (e.g., "don't" to "do not").

NLP Refinement & Stopword Removal

Objective: Isolate high-value keywords by removing low-information linguistic fillers.

Transformation

body_clean → body_no_stopwords

Stopword Dictionary



NLTK Standard List

Utilizes the standard NLTK English stopwords list as the base for linguistic filtering.



Domain-Specific Additions

Custom finance terms filtered to preserve sentiment-heavy signals while removing industry noise.

Finalization & Data Export

Objective: Prepare and serialize the refined dataset for the machine learning modeling phase.



Data Integrity Check

Verified preprocessing steps and validated engineered features to ensure no null values or leakage.



Feature Tokenization

Converted text into sequences and applied vectorization for classification-ready formats.



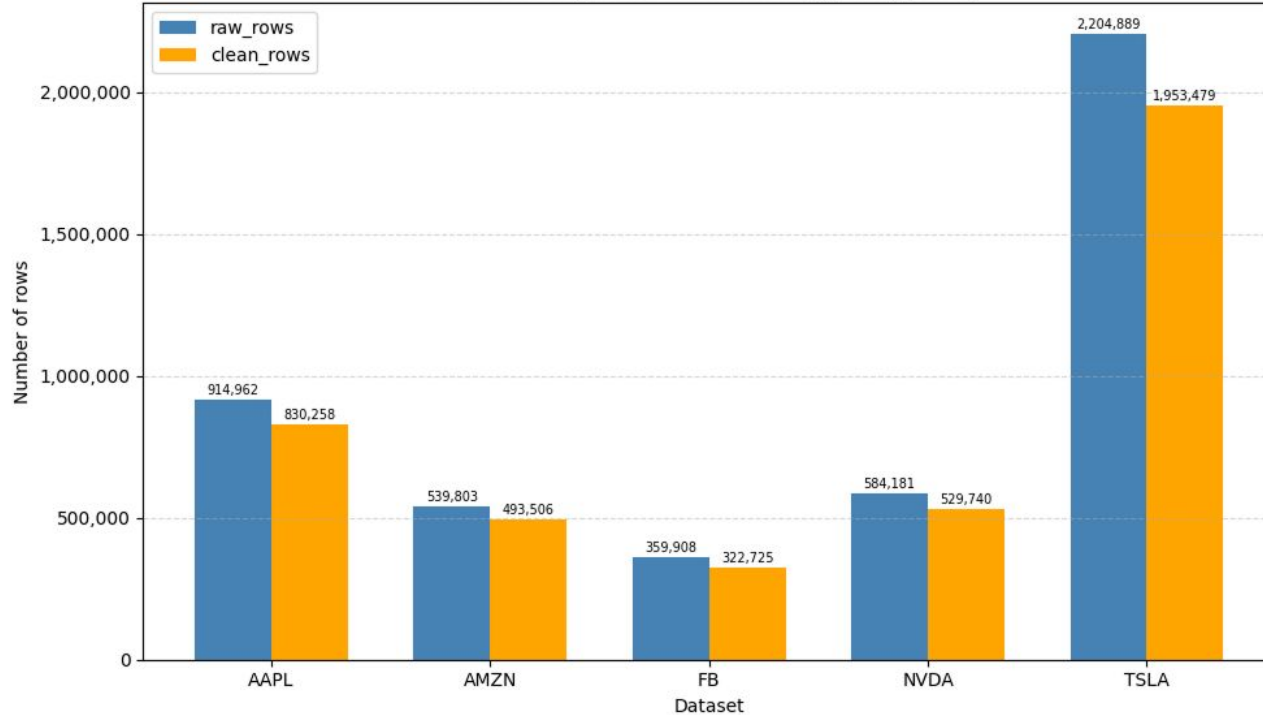
File Generation

Serialized processed ticker datasets (AAPL, AMZN, FB, NVDA, TSLA) as [ticker]_final.csv.

Ready for Modeling Phase

Refined Dataset Exported

Raw vs Deduplicated Row Counts (by body_clean)



Duplicates removed:

AAPL: 914,962 → 830,258
(84,704 removed, 9.3%)

AMZN: 539,803 → 493,506
(46,297 removed, 8.6%)

FB: 359,908 → 322,725
(37,183 removed, 10.3%)

NVDA: 584,181 → 529,740
(54,441 removed, 9.3%)

TSLA: 2,204,889 → 1,953,479
(251,410 removed, 11.4%)